No of Pages :2 **Course Code : 23MX21**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004

SEMESTER EXAMINATIONS, JUNE 2024

MCA Semester: 2

23MX21 SOFTWARE ENGINEERING

Time: 3 Hours **Maximum Marks:100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks each, (b) carries 7 marks each and (c) carries 15 marks each.
3. **COCOMO table may be permitted.**
4. Course Outcome : Table

Qn.1	CO1.	Qn.2	CO2.	Qn.3	CO3.	Qn.4	CO4.

COs	At the end of learning this course, students will be able to
CO1	Analyse software development methodologies and life cycle models and select appropriate ones for the given problem and compute the project estimate.
CO2	Perform requirements gathering and prepare SRS for a given application as per IEEEformat.
CO3	Apply UML for a sample OOAD application and prepare design documents as per IEEE
CO4	Select and apply appropriate testing methodologies for testing different types of software applications and analyse the test results

Question **(BTL)**

1. a) Write your comments on the following statement: (L2)

"Practicing software engineering while developing software will tend you to create voluminous documentation and this will eventually slow down the process of software development"

b) i) For a project beset with many risks, would you recommend the prototyping or the spiral model? Justify the answer. (4) (L4)

ii) List the 4 key characteristics of Agile Software Development. (3) (L1)

c) Consider an Organic project and it is decomposed into 7 distinct functions. Apply decomposition techniques to compute total number of lines of code and cost required to complete the project. Use the following empirical equation to propose the LOC values: C1 + C2F where C1 and C2 are constants and F is the Function number. Use the following constant table to propose three LOC values. Assume that 2% of m will be paid as $/Loc. (L4)

Constants/LOCs	a	m	b
C1	351	373	450
C2	624	692	737

Using the above LOC value, apply COCOMO method to compute other estimates by assuming the following cost drivers. Very high skilled programmer and analyst; High - Data Base size; Low product complexity; High - application experience.

Use a=3.2, b=1.05, c=2.5 and d=0.38.

2. a) Is it true that, prototyping is a technique of building a software which helps in uncovering and capturing software requirements from client? Justify the answer (L4)

b) i) Draw the software Quality Attribute Triangle. (3) (L1)

ii) What are the problems in the following requirement? (L5)

"If sales for current month are below target sales, then report is to be generated unless difference between target sales and actual sales is less than half of difference between target sales and actual sales in previous month, or if difference between target sales and actual sales for the current month is less than 5%" (4)

c) i) 1) Tabulate the functional and non-functional requirements of a system to admit the students in the Post Graduate Degree based on TANCET. (5) (L3)

2) Illustrate the same using a neat USECASE model. (10) (L6)

(OR)

ii) 1) Design a suitable USE CASE Model to depict our college Library automation software [Online Public Access Catalogue]. (8) (L6)

2) Write a complete USE CASE Story to explain the student authentication to access our library. (7) (L4)

3. a) Is it true that if there is an increase in the cohesion; coupling would automatically decrease? Justify the answer. (L2)

b) Design a simple Class diagram for the ATM Banking software (L6)

c) i) Design a neat activity model to explain the different steps in ordering of foods using a standard mobile app. (L6)

(OR)

ii) Design a neat sequence and collaboration diagrams to explain the withdrawal of cash from ATM considering that, the customer is already authenticated. (L6)

4. a) What is the significance of Pareto Principle in software testing? (L2)

b) Apply Black Box method to test a module which is developed for customer authentication in an ATM Banking Software. (L3)

c) i) Write a complete pseudo code to read a number and to display the input number is positive or negative or zero. Repeat the reading of numbers from the user till their response is NO; Output the count on positive, negative and zeros read. (L6)

ii) Apply BASIS PATH testing for this code and present the test case report in the standard template. (L5)

CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL

/END/

FD/RL